

Summary of Prototype

STUDENT user journey

1. Verify email
2. Student can update technical skills and interests

Student Portal

Enter your registered email address to access the portal:

vic@gmail.com

Verified: Welcome back, Vic!

Student ID	Cohort	Email Address	Capstone Status
#1004	4	vic@gmail.com	Matched

Update Your Core Credentials

Technical Skills

Python, Robotics, Web development

Domain Interests

Sensor tech

Save Profile Changes

3. Look through list of Available Projects and click apply

ProjectsMy ApplicationsSubmit Self-Sourced Project

Available Approved Projects

ML classification of Chest X-Rays

Location: Hybrid | Duration: 12 months | Available Places: 2

Using ML to classify chest X-ray images

Required Skills: Python, ML

Apply

X-ray to Text

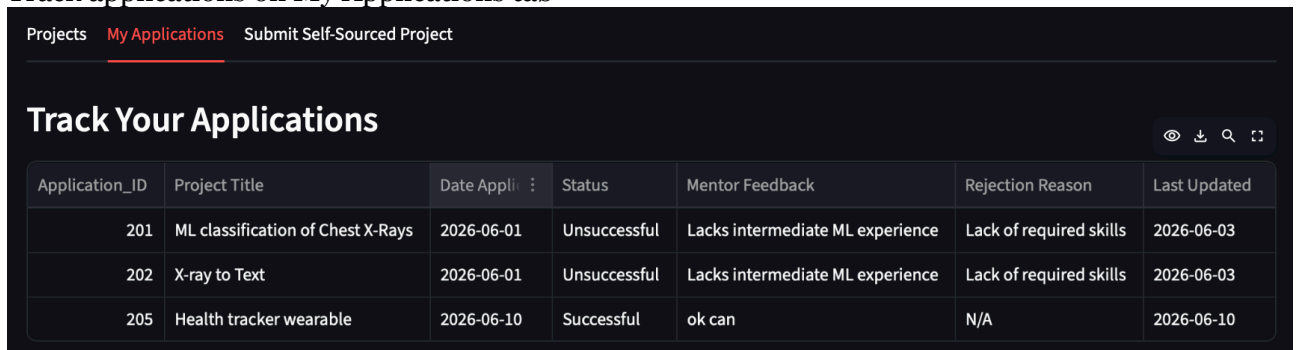
Location: Hybrid | Duration: 12 months | Available Places: 1

Image parsing to translate x-ray images to text description

Required Skills: Python, ML

Apply

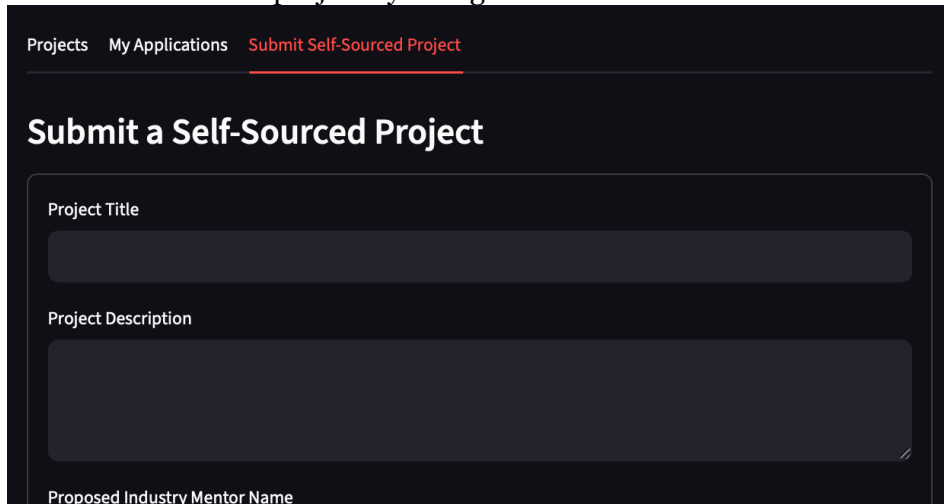
4. Track applications on My Applications tab



The screenshot shows the 'My Applications' tab with a sub-tab 'Submit Self-Sourced Project'. The main heading is 'Track Your Applications'. Below it is a table with 7 columns: Application_ID, Project Title, Date Applied, Status, Mentor Feedback, Rejection Reason, and Last Updated. There are three rows of data.

Application_ID	Project Title	Date Applied	Status	Mentor Feedback	Rejection Reason	Last Updated
201	ML classification of Chest X-Rays	2026-06-01	Unsuccessful	Lacks intermediate ML experience	Lack of required skills	2026-06-03
202	X-ray to Text	2026-06-01	Unsuccessful	Lacks intermediate ML experience	Lack of required skills	2026-06-03
205	Health tracker wearable	2026-06-10	Successful	ok can	N/A	2026-06-10

5. Submit self-sourced project by filling in the form



The screenshot shows the 'Submit Self-Sourced Project' form. It has three main input fields: 'Project Title' (a single-line text box), 'Project Description' (a multi-line text area), and 'Proposed Industry Mentor Name' (a single-line text box).

Applying for a Self sourced project

1. Student fills in form and submits
2. Admin reviews and approves / rejects / rejects and request for resubmission with more details
 1. Approve: Student's capstone status is now Matched, project's approval status is Approved, application status is Successful
 2. Reject: Project's approval status is Rejected, potentially with comments on reasons for rejection

MENTOR user journey

1. Verify email

- Submit new project by filling in the form

Mentor Portal

Enter your registered mentor email address to unlock your dashboard:

robbie@abc.com

Verified: Welcome back, Robbie!

Mentor ID

Company

Role / Department

#5001

ABC Health ...

Principal Re...

Submit New Project

Review Student Applications

Offer a New Capstone Project

Submit a standard project description to be added to the student marketplace upon admin vetting.

Project Title

Project Description

- Admin will approve the project or reject and can include a message on reasons for rejecting
- Mentor sees list of projects they have submitted and admin statuses

Your Projects

This table tracks projects you have proposed

Project ID	Project Title	Location	Duration	Admin Status
101	ML classification of Chest X-Rays	Hybrid	12 months	Approved
102	X-ray to Text	Hybrid	12 months	Approved

- Mentor can review all student applications to their projects.

Submit New Project

Review Student Applications

All Student Applications

Below is a master list of all students who have applied to your capstone projects.

Application_ID	Project Title	Applicant Student	Project Admin Status	Application Match Status	Date Applied
201	ML classification of Chest X-Rays	Vic	Approved	Unsuccessful	2026-06-01
202	X-ray to Text	Vic	Approved	Unsuccessful	2026-06-01
203	ML classification of Chest X-Rays	Mel	Approved	Successful	2026-06-04
204	X-ray to Text	Frank	Approved	Submitted	2026-06-04

6. AI summary helps to summarise student profile and the fit with a specific project

AI-Generated Candidate Fit Summary

Automated analysis comparing the applicant's profile text against your project requirements.

[Run AI Profile Review](#)

Fit Analysis: The applicant demonstrates a strong academic background in data structures and Python programming, aligning well with your technical stack.

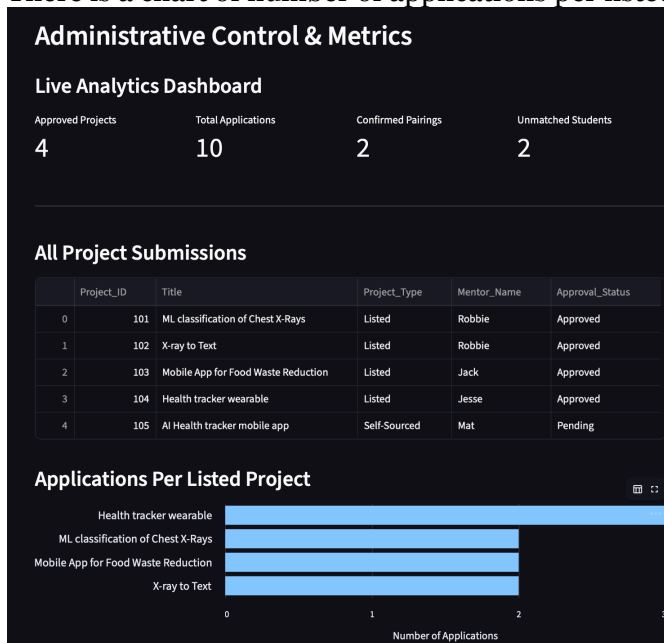
Key Alignment Markers:

- Student Statement:** Highly interested in automated pipeline architectures and machine learning model tracking.
- Project Match:** Strong crossover with your project's focus on structured data parsing. The student's elective coursework in predictive modeling directly supports your milestone goals.
- Recommendation:** High Fit. The student possesses the foundational coding skills required, though they may need initial mentorship regarding domain-specific corporate workflows.

7. Mentor approves or declines candidates for their projects

ADMIN user journey

1. Admin can see number of approved projects, total applications, confirmed pairings, unmatched students
2. There is a table of all project submissions from either mentors or students self-sourced
3. There is a chart of number of applications per listed project



4. There are tables of successful matches and unmatched students. Unmatched students who suffered at least 2 rejections are highlighted to the admin.

Pairing Status Overview

Successful Matches

Pairing ID	Student ID	Student Name	Mentor ID	Mentor Name	Project Title
1	1001	Mel	5001	Robbie	ML classification of Chest X-Rays

Support Needed: Student Vic (#1004) has suffered 2 rejections and is still Unmatched. Immediate intervention recommended.

Unmatched Students

All registered candidates who have not yet secured a finalized project allocation match.

Student ID	Student Name	Cohort	Current Status	Total Rejections
1004	Vic	4	Unmatched	2
1002	Frank	4	Unmatched	0
1003	Sam	4	Unmatched	0

5. There is a table of common rejection reasons.

Common Reasons for Unsuccessful Applications	
Stated Reason / Feedback Category	Count
Resubmit with more details (Scope unverified)	3
Project timeline does not meet capstone duration criteria	2
Missing explicit corporate mentor sign-off credentials	1
Technical scope too narrow	1
Duplicate project proposal submission detected	1

6. Project Approvals Pipeline: admin uses this to approve or reject project submissions from students or mentors

Project Approvals Pipeline

Pairing Approvals & Letter Generation

Review Submitted Projects (Listed & Self-Sourced)

	Project_ID	Title	Mentor_Name	Mentor_Email	Project_Type	Approval_Status
4	105	AI Health tracker mobile app	Mat	mat@aihealth.com	Self-Sourced	Pending

Choose a Project ID to take action on:

105

Action:

☐ Approve

☒ Reject

Enter Rejection Reason / Clarification notes:

Resubmit with more details

Process Project

7. Pairing Approvals & Letter Generation: Admin can generate letter to send to student/mentor for confirmation of project.

Project Approvals Pipeline

Pairing Approvals & Letter Generation

Finalize Matches & Generate Draft Letters

	Application_ID	Student_Name	Title	Company	Mentor_Name
0	203	Mel	ML classification of Chest X-Rays	ABC	Robbie
1	205	Vic	Health tracker wearable	NTU	Jesse

Select Application ID to generate letter:

203

Run Rule-Check & Generate Confirmation Letter

Generated Draft (Ready to copy/send):

DEAR MEL,

We are pleased to inform you that your application for the Capstone Project "ML classification of Chest X-Rays" has been formally approved.

You will be operating under the supervision of Robbie at ABC.

Please coordinate directly with your mentor regarding onboarding timelines.

Data

Sample data for the 5 tables: students, mentors, projects, applications, pairings.

Students

	Student_ID	Student_Name	Student_Email	Cohort	Skills	Interests	Capstone_Status
0	1001	Mel	mel@gmail.com	3	Python, SQL, Tableau, ML	Computer Vision	Matched
1	1002	Frank	frank@gmail.com	4	Python, Computer Vision, ML	Computer Vision	Unmatched
2	1003	Sam	sam@gmail.com	4	Java, Python, Web development, Mobile App Dev	App Dev	Unmatched
3	1004	Vic	vic@gmail.com	4	Python, Robotics, Web development	Sensor tech	Unmatched

Mentors

	Mentor_ID	Mentor_Name	Mentor_Email	Company	Department_or_Role
0	5001	Robbie	robbie@abc.com	ABC Health Lab	Principal Researcher
1	5002	Jack	jack@abc.com	ABC Enterprise	Engineering Lead
2	5003	Jesse	jesse@abc.com	ABC Innovation Lab	Director
3	5004	Mark	mark@lumon.com	Lumon	External Advisor

Projects

	Project_ID	Mentor_ID	Mentor_Name	Mentor_Email	Company	Title	Description	Project_Type
0	101	5001	Robbie	robbie@abc.com	ABC Health Lab	ML classification of Chest X-Rays	Using ML to classify chest X-ray images	Listed
1	102	5001	Robbie	robbie@abc.com	ABC Health Lab	X-ray to Text	Image parsing to translate x-ray images to text	Listed
2	103	5002	Jack	jack@abc.com	ABC Enterprise	Mobile App for Food Waste Reduction	Building a cross-platform application	Listed
3	104	5003	Jesse	jesse@abc.com	ABC Innovation Lab	Health tracker wearable	Creating a wearable device to track health metrics	Listed
4	105	5004	Mark	mark@lumon.com	Lumon	Drones	Drones for surveying terrains	Self-Sourced
5	106	5005	Mark	mark@lumon.com	Lumon	Automated Drones for tough terrains	Drones for helping humans with dangerous tasks	Self-Sourced

Project_Type	Required_Skills	Location	Duration	Stipend	Available_Places	Approval_Status	Open_Full_Status	Admin_Comments
Listed	Python, ML	Hybrid	12 months	\$1000	2	Approved	Open	NA
Listed	Python, ML	Hybrid	12 months	\$1000	1	Approved	Open	NA
Listed	Web development, Mobile App Dev	Hybrid	12 months	\$800	2	Approved	Open	NA
Listed	Sensor tech, Embedded Systems	On-site	6 months	\$300	1	Approved	Open	NA
Self-Sourced	Python	On-site	3 Months	\$300	1	Rejected	Open	Resubmit with more details
Self-Sourced	Python, Robotics	On-site	3 Months	\$300	1	Approved	Open	NaN

Applications

Application_ID	Student_ID	Project_ID	Date_Applied	Application_Status	Mentor_Comments	Reason_Unsuccessful	Date_Updated
0	201	1004	101 2026-06-01	Unsuccessful	Lacks intermediate ML experience	Lack of required skills	2026-06-03
1	202	1004	102 2026-06-01	Unsuccessful	Lacks intermediate ML experience	Lack of required skills	2026-06-03
2	203	1001	101 2026-06-04	Successful	Ok	NA	2026-06-04
3	204	1002	102 2026-06-04	Submitted	NA	NA	2026-06-05
4	205	1004	105 2026-06-10	Successful	NA	NA	2026-06-10
5	206	1003	104 2026-06-10	Unsuccessful	NA	Insufficient experience in health tech	2026-06-10
6	207	1003	103 2026-06-10	Successful	Welcome	NA	2026-06-10

Pairings

Pairing_ID	Student_ID	Mentor_ID	Project_ID	Project_Type	Title	Start_Date	Duration	Stipend	Approval_Status
0	1	1001	5001	101 Listed	ML classification of Chest X-Rays	2026-07-01	12 months	\$1000	Approved
1	2	1004	5005	106 Self-Sourced	Automated Drones for tough terrains	2026-06-10	3 Months	\$300	Approved

Automation/AI features

- Rule-based flagging of students who have had multiple rejections and still unmatched to a project, allowing Admin to quickly identify them and provide support if necessary.
- AI summary of student's fit with a project will reduce the amount of time and effort required to read through student's profiles and multiple projects, helping the mentors assess the fit more easily. Students profiles may contain a lot more information than the dummy data, or statements of interest. AI is not used for making the decision for the mentor, as it is only a tool that summarises large amounts of text.
- Auto generation of confirmation message that the Admin can copy and send to students. Admin still needs to review the message before sending it manually. The text can be modified to fit specific situations if needed.

Data Privacy and Governance

- Role-based access for students: Granted execution rights to view and apply projects, submit self-sourced projects, update credentials exclusively to each student's email. They do not have visibility of other student's details.
- Role-based access for mentors: Authorised to view applications specific to their active project IDs. Mentors can submit new projects, execute evaluations, and view profiles of students who applied to their projects. They cannot view projects hosted by others or access the admin functions.

- Admins has access to all students, projects and applications, has rights to reject or approve projects or pairings, and generate formal paperwork.
- Audit Trail: Aside from confirmed and successful applications, unsuccessful or rejected applications are also logged with unique IDs for easy tracing.
- Dummy Data: for the purpose of demonstration, the data used is synthetic

Further work and additional features

- Password authentication
- After a student gets one successful matching, other applications made by the student should be automatically rejected.
- Remove a project from the list on Student portal after the maximum openings have been taken up and confirmed

Prototype-to-Production

Tech Stack

- Compute layer: Python for executing the core matching logic, data processing pipelines, and data transformations.
- Interface layer: Streamlit for front-end UI, linked to production database instead of volatile in-memory states
- Database: PostgreSQL to ensure permanent data persistence
- AI: Ollama API open models or cloud models

Key Phases

1. Database Migration from pandas to permanent SQL tables and rewrite Python access methods
2. Optimise LLM output by refining system prompts
3. Run user acceptance testing with trail data, stress test with hundreds of student profiles

Risks

- Model hallucinations for summaries: AI will only serve as a summary tool, and the mentor still has access to original student profiles. Mentor has to make the decisions manually.

Support

- Compute infrastructure to host AI models such as AWS servers, to handle text-processing workloads with minimal latency.
- Support from institutional IT administrators to securely connect and migrate real student and mentor data profiles into the production PostgreSQL database.